MATH 2228-601, SUMMER II 2025, MOCK FINAL

INSTRUCTIONS

1. Do not open the exam paper or start working until you are told to do so.

2. Bubble in your name as DOE John S (if it is John S. Doe), bubble in your student ID number.

3. NO NOTES OR BOOKS ARE ALLOWED. You may not communicate with others during the exam. Please be considerate and refrain from making noise. Switch off any beepers, cell phones, etc.

4. Bubble in your answer to, say, Question 4, on line 4.

5. The exam is self-explanatory. The instructor will not interpret the questions for you. You may, however, ask for another copy of the exam if your copy is illegible.

6. Have your calculator in **RADIAN MODE**. You can set radian mode by pressing a button called MODE, SET UP, or DRG (degree, radian, gradian).

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1. Let Q_1 be the mean of the numbers given by f_i copies of x_i listed below as pairs (x_i, f_i).

$(65, 54)$,$(-82, 66)$,$(-15, 85)$,$(98, 90)$,$(-22, 69)$,$(-39, 60)$,$(43, 64)$,$(-38, 44)$,$(-59, 28)$,$(-16, 17)$

Let $Q = \ln(3 + |Q_1|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

2. Let Q_1 be the sample mean and Q_2 the sample standard deviation of the numbers given by f_i copies of x_i listed below as pairs (x_i, f_i).

$(81, 70)$,$(-27, 36)$,$(46, 76)$,$(37, 69)$,$(4, 76)$,$(-66, 79)$,$(-56, 36)$,$(-95, 48)$,$(-36, 38)$,$(-86, 21)$

Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

3. Let Q_1 be the sample mean and Q_2 the sample standard deviation of the numbers listed below.

12.88, -8.52, -6.83, 8.72, 13.63, 0.05, -5.55, 17.34, 18.83, 0.63, 18.21, 3, -2.01, 12.9, 1.69, 1.02, -1.44, 19.19, -2.78, -0.09

Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

4. Let Q_1 be the weighted GPA (grade point average) rounded to two decimals after the point of a college student for the semester given the classes below. Note that A corresponds to 4 points, B to 3, C to 2, D to 1, F to 0.

Class 1, grade B, credit 4; Class 2, grade A, credit 3; Class 3, grade F, credit 3; Class 4, grade C, credit 2; Class 5, grade D, credit 5

Let $Q = \ln(3 + |Q_1|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

5. Let Q_1 be the sample mean rounded to two decimals after the point and Q_2 the sample standard deviation rounded to two decimals after the point of the salary at a company where 278 people earn in the bracket $(30000, 45000]$, 262 people earn in the bracket $(45000, 58000]$, 249 people earn in the bracket $(58000, 68000]$, 238 people earn in the bracket $(68000, 77000]$, 223 people earn in the bracket $(77000, 96000]$, which accounts for everybody. Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

6. Suppose a distribution has mean $\mu = 341$ and standard deviation $\sigma = 51.1$. Let Q_1 be the value which belongs to the z-score -1.2, Q_2 the value which belongs to the z-score -1.8, Q_3 the z-score of the value 187.7, and Q_4 the z-score of the value 203.03. Let $Q = \ln(3+|Q_1|+2|Q_2|+3|Q_3|+4|Q_4|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

7. Let Q_1 be the minimum, Q_2 the first quartile, Q_3 the median, Q_4 the third quartile, and Q_5 the maximum of the list below.

152, 689, 608, 717, 688, 857, 469, 318, 127, 559, 610, 661, 850, 633, 322, 469, 391, 447, 559, 828, 782, 160, 424

Let $Q = \ln(3+|Q_1|+2|Q_2|+3|Q_3|+4|Q_4|+5|Q_5|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

8. Let Q_1 be the slope, Q_2 the intercept of the linear regression line $y = ax + b$, and Q_3 the prediction $\hat{y}_0 = ax_0 + b$ for $x_0 = 20.77$, where the sequences x and y are as follows:

x: $94, -83, 15, -85, 22, 82, 10, -19, 21, -57, 57, 92,$

y: $52, 45, -7, 84, -34, -49, -82, -42, 95, 17, -84, -54.$

Let $Q = \ln(3+|Q_1|+2|Q_2|+3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

9. The table below shows how many parts in a lot fail or do not fail when made of one of three metal alloys. Relative to the table below, let Q_1 be the probability that a part fails given that it is made of metal A, Q_2 the probability that a part fails, Q_3 the probability that a part does not fail given that it is not made of metal B, and Q_4 the probability that a part fails or it is made of metal C.

	metal A	metal B	metal C
part fails	182	230	168
part does not fail	505	783	352

Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3| + 4|Q_4|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

10. Suppose that the events E, F, G are independent and have probabilities $P(E) = 0.258$, $P(F) = 0.292$, $P(G) = 0.11$. Denote the complement of an event H by H^c. Let $Q_1 = P(E \cap F^c \cap G)$, $Q_2 = P(E^c \cap G)$, and $Q_3 = P(E \cap F \cap G^c)$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

11. Let Q_1 be the number of all the ways in which we can select a committee consisting of a chairman, a vice chairman, a treasurer, and 10 ordinary members (whose order does not matter) out of 25 people. Let $Q = \ln(3 + |Q_1|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

12. Let X be the number of successful rocket launches in 636 independent trials with success probability 0.927. Using the binomial distribution, let Q_1 the probability that $X \geq 587$, Q_2 the probability that $X \leq 596$, and Q_3 the probability that $584 \leq X \leq 599$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

13. Let X be a normal random variable with mean 258 and standard deviation 14.82. Let Q_1 the probability that $X \geq 241$, Q_2 the probability that $X \leq 291$, and Q_3 the probability that $229 \leq X \leq 267$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

14. Let $X_1, X_2, \ldots, X_{19}$ be independent normally distributed random variables with mean -28 and standard deviation 1.4. Let $\bar{X} = \frac{1}{19}(X_1 + X_2 + \ldots + X_{19})$. Let Q_1 be the mean of $\bar{X}$ and Q_2 the standard deviation of $\bar{X}$. Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

15. Using the normal distribution that approximates the sampling distribution of the population proportion if the size of the population is 18684, the sample size is 519, and the success probability is 0.876, let $Q_1 = P(\hat{p} < 0.836)$, $Q_2 = P(\hat{p} > 0.852)$, and $Q_3 = P(0.909 < \hat{p} < 0.919)$, where $\hat{p}$ is the sample proportion. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

16. Let Q_1 be the lower and Q_2 the upper end point of the 93 percent confidence interval for the sample proportion $\hat{p}$ if the success count in the sample is $x = 144$ and $n = 516$ is the sample size. Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. —

(E) $4 \leq T \leq 5$.

17. Given the numbers below drawn from a normal distribution, let Q_1 be the lower and Q_2 the upper end point of the 94 percent confidence interval for the sample mean.

93.01, 90.89, 92.08, 92.09, 91.81, 92.61, 91.89, 94.22, 92.66, 92.87, 91.88, 93.04, 91.44, 92.34, 90.57

Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

18. Test at the 91 percent level of significance the null hypothesis $H_0\colon p = 0.572$ versus the alternative hypothesis $H_1\colon p > 0.572$, where p is the population proportion, $n = 564$ is the sample size, and $x = 340$ is the number of observed "successes". Let $Q_1 = \hat{p}$ be the sample proportion, Q_2 the z-statistic, and $Q_3 = 1$ if we reject the null hypothesis H_0, and $Q_3 = 0$ otherwise. Let $Q = \ln(3+|Q_1|+2|Q_2|+3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

19. Test at the 91 percent level of significance the null hypothesis $H_0\colon p = 0.429$ versus the alternative hypothesis $H_1\colon p \neq 0.429$, where p is the population proportion, $n = 796$ is the sample size, and $x = 381$ is the number of observed "successes". Let $Q_1 = \hat{p}$ be the sample proportion, Q_2 the z-statistic, and $Q_3 = 1$ if we reject the null hypothesis H_0, and $Q_3 = 0$ otherwise. Let $Q = \ln(3+|Q_1|+2|Q_2|+3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

20. Relative to a normal population, test at the 95 percent level of significance the null hypothesis $H_0\colon \mu = 470$ versus the alternative hypothesis $H_1\colon \mu \neq 470$, where μ is the population mean, $n = 10$ is the sample size, $\bar{x} = 467.85$ is the sample mean, and $s = 2.99$ the sample standard deviation. Let Q_1 be the degrees of freedom for the t-distribution, Q_2 the t-statistic, and $Q_3 = 1$ if we reject the null hypothesis H_0, and $Q_3 = 0$ otherwise. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.